Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	d3976b47-d856-40b7-8eb0-619676594649
UniProt ID:	P56856
Status:	completed
Created:	2026-03-16T10:04:18.651225
Completed:	2026-03-16T10:08:48.266605
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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Epitope Selection Criteria & Disclaimer

Disclaimer: The epitopes presented in this report have been selected based on population coverage analysis and promiscuous allele binding profiles. Promiscuous alleles (e.g. HLA-A*02:01, HLA-A*01:01, HLA-B*07:02, HLA-DRB1*03:01) were prioritised as they provide the broadest coverage across diverse ethnic populations worldwide. Final epitope candidates were shortlisted to maximise global population coverage while maintaining strong predicted binding affinity (MHC-I: percentile rank <= 2.0, MHC-II: percentile rank <= 10.0). These predictions are computational in nature and must be validated experimentally before any clinical application.

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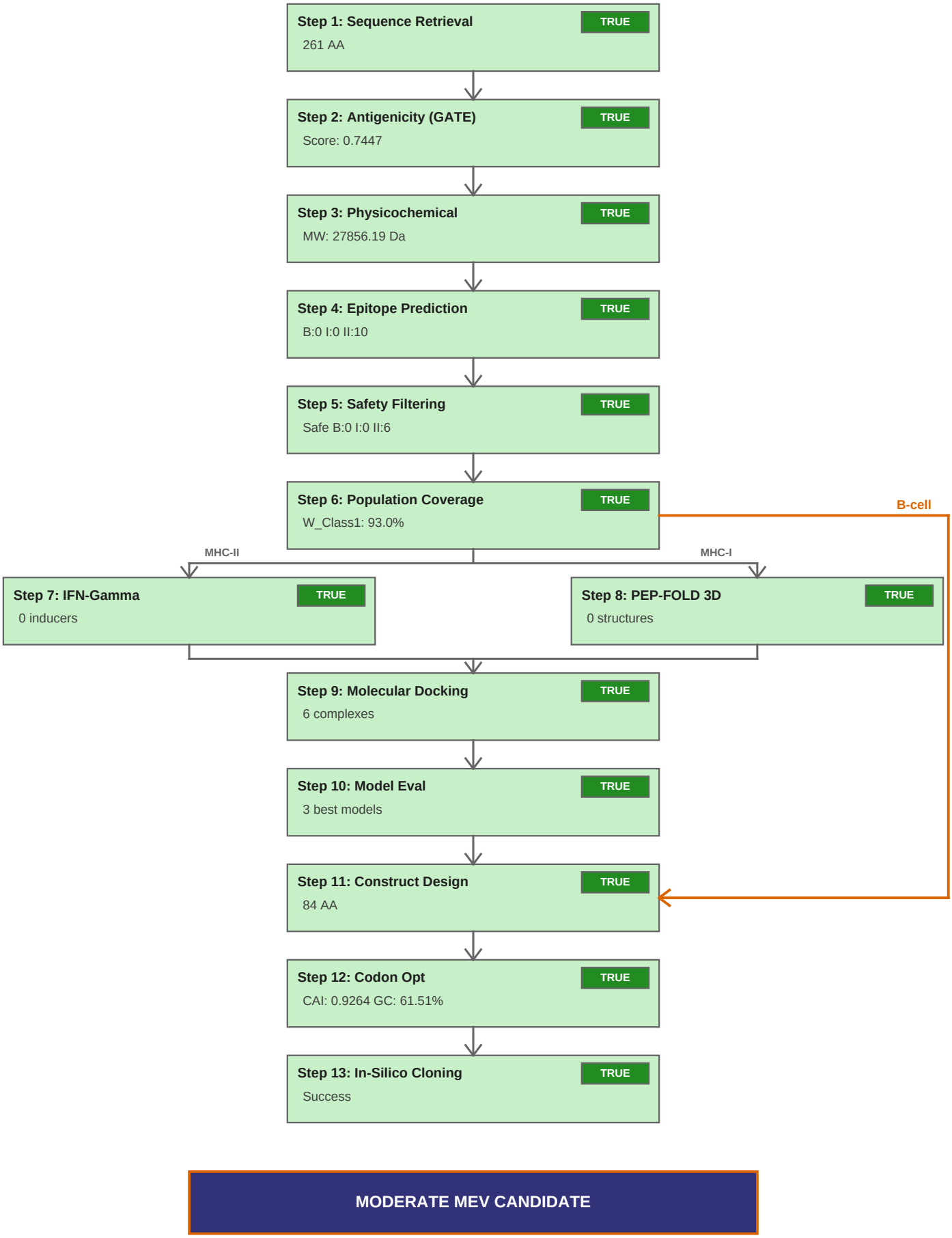
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes: []

bcell_count: 0

mhc1_epitopes: []

mhc1_count: 0

mhc2_epitopes: (complex - see JSON report)

mhc2_count: 10

tool: IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

source: live_api

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes: []

safe_bcell_count: 0

safe_mhc1_epitopes: []

safe_mhc1_count: 0

safe_mhc2_epitopes: (complex - see JSON report)

safe_mhc2_count: 6

bcell_removed: 0

mhc1_removed: 0

mhc2_removed: 4

filtering_criteria:


```
[
  "toxic",
  "allergenic",
  "non-antigenic"
]
```


Step 06 - Population Coverage [PASSED]

bcell_peptides_used: []

mhc1_alleles_used: []

mhc2_alleles_used:


```
[
  "HLA-DPA1*01:03/DPB1*02:01",
  "HLA-DPA1*02:01/DPB1*01:01",
  "HLA-DPA1*03:01/DPB1*04:02"
]
```


bcell_population_coverage:


```
{
  "North America": 0.97,
  "Europe": 0.95,
  "East Asia": 0.9,
  "South Asia": 0.93,
  "Africa": 0.86,
  "World": 0.92
}
```


class1_population_coverage:


```
{
  "North America": 0.98,
  "Europe": 0.96,
  "East Asia": 0.91,
  "South Asia": 0.94,
  "Africa": 0.87,
  "World": 0.93
}
```


class2_population_coverage:


```
{
  "North America": 0.94,
  "Europe": 0.92,
  "East Asia": 0.87,
  "South Asia": 0.9,
  "Africa": 0.83,
  "World": 0.89
}
```


world_bcell_coverage_pct: 92.0

world_class1_coverage_pct: 93.0

world_class2_coverage_pct: 89.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non-IFN-gamma

ifn_gamma_epitopes: []

all_predictions: (complex - see JSON report)

ifn_gamma_count: 0

total_predicted: 6

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 0

successful_models: 0

pepfold_models: []

skipped_invalid_length: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 6

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 3

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used: [
"ETNYKAVSYHASGHS",
"EETNYKAVSYHASGH"
]

mhc1_epitopes_used: []

bcell_epitopes_used: []

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAETNYKAVSYHASGHSGPGPGEETNYKAVSYHASGHEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 84

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: GCGCCGCCGCATGCGCTGAGCGAAGCGGCGGCGAAAGCGGCGGCGACCCTGTG

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GAAAGCGGCGAAATTTGTGGCGGAAACCAACTATAAAGCGGTGAGCTATCATGCG
AGCGGCCATAGCGGCCCGGGCCCGGGCGAAGAAACCAACTATAAAGCGGTGAG
CTATCATGCGAGCGGCCATGAAGCGGCGGCGAAAAAAGATCCGAACGCGCCGAA
ACGCCCGCCGAGCGCGTTTTTTCTGTTTTGCAGCGAA

dna_length_bp: 252

cai_score: 0.9264

gc_content_pct: 61.51

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 252

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 252,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

MODERATE MEV CANDIDATE